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Review

Hybrid Laser Additive Manufacturing of Metals: A Review

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Abstract: Due to the unparalleled benefits of traditional processing techniques, additive manufacturing technology has experienced rapid development and continues to expand its applications. However, as industrial standards advance, the pressing needs for high precision, high performance, and high efficiency in the manufacturing sector have emerged as critical bottlenecks hindering the technology's progress. Single-laser additive manufacturing methods are insufficient to meet these demands. This review presents a comprehensive exploration of metal hybrid laser additive manufacturing technology, encompassing various aspects, such as multi-process hybrid laser additive manufacturing, additive–subtractive hybrid manufacturing, multi-energy hybrid additive manufacturing, and multi-material hybrid additive manufacturing. Through a thorough examination of the principles of laser additive manufacturing technology and the concept of hybrid manufacturing, this paper investigates in depth the notable advantages of hybrid laser additive manufacturing technology. It provides valuable insights and recommendations to guide the development and research of innovative machining technologies.

Keywords: laser additive manufacturing; hybrid manufacturing; multi-process; metal

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1. Introduction

Additive manufacturing technology (AM), colloquially known as three-dimensional (3D) printing, is a manufacturing technique in which metallic things are constructed layer by layer from a computer model [1]. It offers unparalleled benefits over conventional fabrication techniques in that it can generate components with complicated architectures, such as meshes and voids [2], and facilitate the synthesis of functionally gradient materials [3]. It is deployed in several aeronautical, medicinal, automotive, and marine applications [4–7]. Using a laser as an energy source, laser additive manufacturing is a non-equilibrium metallurgical process involving a variety of complicated material metallurgy and physical and thermal coupling phenomena [8–10]. Surface defects such as porosity, spheroidization, unmelting, delamination, and fractures are prevalent, reducing the surface quality of produced components and reducing the mechanical properties, such as fatigue strength [11]. With more than 30 years of rapid development, additive manufacturing technology advances toward intelligence and industrialization [12].

When laser additive manufacturing is combined with other techniques, a hybrid laser additive manufacturing technology emerges. It has the potential to broaden the range of industrial applications, achieve high efficiency in the production of high-precision and high-performance components, and overcome the limits of laser additive manufacturing technology in terms of poor surface integrity and metallurgical flaws.

This review meticulously examines metal hybrid laser additive manufacturing technologies, leveraging the fundamental principles of laser additive manufacturing (LAM) in conjunction with the concept of hybrid manufacturing. Guided by the theoretical framework of PMEH (property–mechanism–energy source–hybrid-AM process), our analysis emphasizes the intricate relationships among various hybrid machining technologies and

their collective impact in improving component quality, functionality, and mechanical properties. We carry out our work by systematically categorizing and analyzing diverse hybrid processing methodologies, including multi-process hybrid laser additive manufacturing, additive–subtractive hybrid manufacturing, multi-energy hybrid additive manufacturing, and multi-material hybrid additive manufacturing.

2. Laser Additive Manufacturing Technology

Laser additive manufacturing (LAM) has developed rapidly, and numerous process categories have emerged. It can rebuild the geometrical features of parts and restore their original mechanical properties through layer-by-layer deposition. The cladding material and a portion of the substrate are co-melted and rapidly cooled to form a high-quality coating with excellent metallurgical bonding ability on the substrate surface. The advantages of laser additive manufacturing technology include high processing efficiency, minimal coating dilution and strong bonding strength with the substrate, a high degree of automation, environmental friendliness, and the diversification of cladding powder alloys. Laser powder bed fusion (LPBF) and laser-directed energy deposition (LDED) are presently regarded as the two most versatile AM processes.

2.1. Laser Powder Bed Fusion (LPBF)

In LPBF technology, the laser is used to selectively melt or sinter metal powder. After melting a layer of powder, the platform moves downward, and the scraper pushes the powder to the platform to scan the next layer. Figure 1 depicts the processing principle. This technology includes selective laser melting (SLM) and Selective Laser Sintering (SLS).

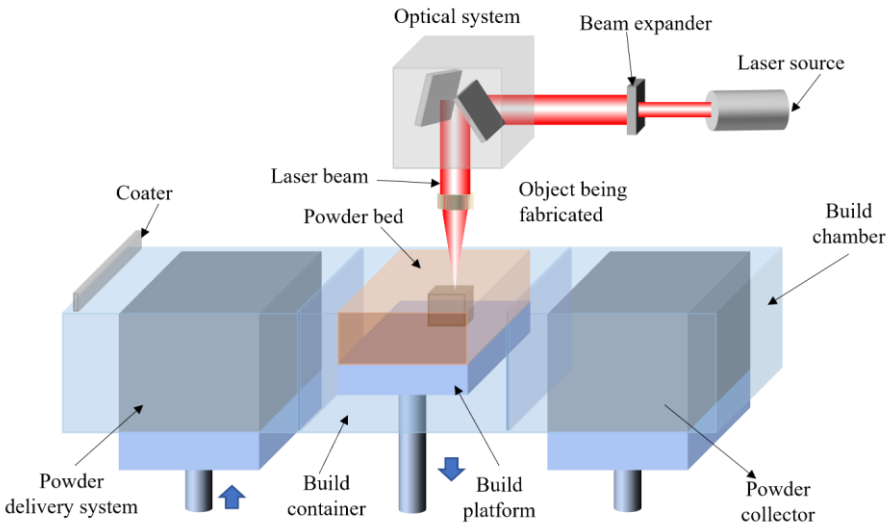


Figure 1. Schematic diagram of laser powder bed fusion (LPBF).

In LPBF technology, the laser selectively melts or deposits metal powder in an inert gas environment, forming a molten pool in and around the substrate that is subsequently swiftly solidified. To reduce oxidation, an inert gas like argon or nitrogen is supplied to the chamber during the laser scanning operation [13]. After melting a layer of powder, the stage descends, and a squeegee pushes the powder onto the stage to scan the next layer. This process is performed multiple times until the component is complete. Figure 1 depicts the LPBF processing principle diagram. The rapid mobility of the laser beam causes the melt pool to cool at rates of up to 106–107 K/s [14]. As a result, it has a considerable advantage in the production of fine microstructures and strong metallurgical bonding capabilities [14,15].

The process parameters significantly affect the melt pool's stability during SLM processing [15]. The width and depth of the melt pool are determined by the laser power and scanning speed. Slower scanning speeds prompt more heating to concentrate in the

melt pool and more material to melt, allowing for the development of bigger and broader melt pools [16]. At the same time, the scanning strategy has a massive influence on the temperature history, residual stresses, and microstructure evolution throughout processing [17,18]. Single-direction raster scanning (SDRS) and cross-direction raster scanning (CDRS) are two popular scanning technologies [19]. Island scanning has the potential to minimize residual stresses [20,21]. Furthermore, the scanning strategy adopted might tailor the morphology and crystallographic texture of the tissue to generate a more anisotropic or isotropic fraction [22].

In comparison to direct energy deposition methods, LPBF can achieve recycled powder. This will considerably decrease material waste and is highly beneficial to environmental conservation and long-term sustainability [15]. When recycled, non-laser-irradiated recycled powders can obtain mechanical characteristics comparable to those of virgin powders [23].

2.2. Laser-Directed Energy Deposition (LDED)

LDED technology melts the powder and deposits it layer by layer on the surface of the substrate using a laser or wire material. In the interaction zone between the laser and the material, a molten pool will form. To reduce the oxidation of the metal surface, an inert gas will be transported to the deposition area [4,24]. Figure 2 depicts the principle in schematic form.

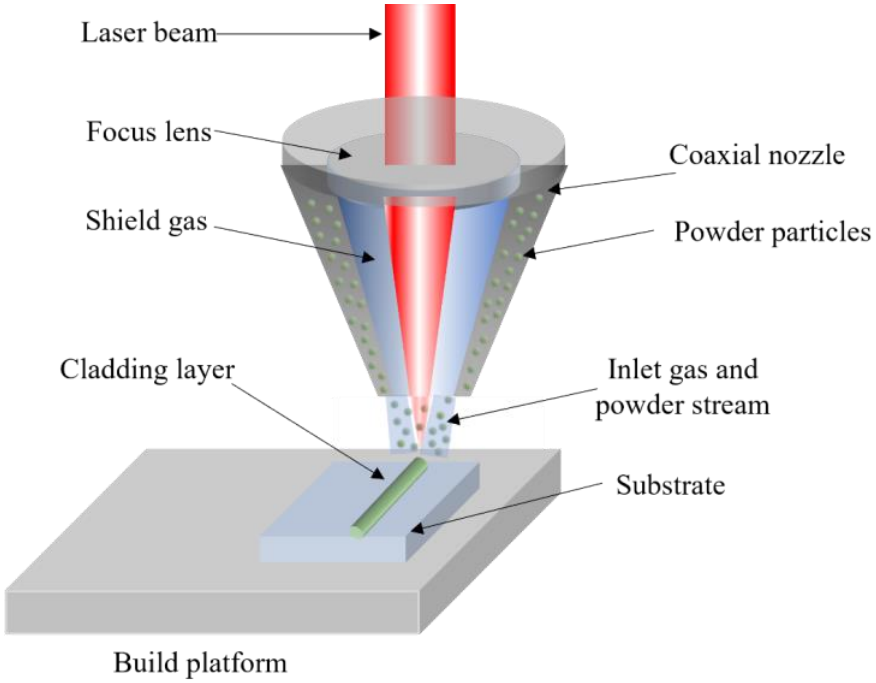


Figure 2. Schematic diagram of laser-directed energy deposition (LDED).

The LDED technique is also known as laser metal deposition (LMD), laser direct metal deposition (LDMD), laser solid forming (LSF), and laser-engineered net shaping (LENS) [25]. LDED is ideal for the deposition of a broad variety of materials, including stainless steel, tool steel, alloy steel, titanium-based alloys, cobalt-based alloys, nickel-based alloys, aluminum alloys, high-entropy alloys, intermetallic compounds, shape memory alloys, ceramics, and composites [4].

LDED enables the in situ alloying of a wide range of alloys and composites [26], along with the adjustment of the powder content during processing to accomplish progressive changes in the chemical composition or microstructure of the component for superior properties. Furthermore, by virtue of its fast build speed, which allows for larger build volumes with less deformation and excellent metallurgical bonding capabilities, this

technology is frequently used to manufacture large near-net-formed parts with favorable mechanical characteristics, such as turbine blades, complete machine impeller discs, and housings [27,28].

3. Hybrid Additive Manufacturing Technology

The International Institute of Production Engineering (CIRP) defines hybrid manufacturing as "a technique based on the interaction of several process mechanisms/energy/tools in the same processing area and at the same time, which has a significant effect on the process performance of parts" [29]. Hybrid manufacturing can be divided into three categories: (1) hybrid process, (2) hybrid equipment, and (3) hybrid materials, structures, or functions.

Recent publications define hybrid-AM processes as combining AM with one or more secondary processes or energy sources that are completely linked and synergistically affect component quality, functionality, and/or process performance [29]. The limitations of a single production method make it tricky to ensure that a product will perform as expected. Advances in hybrid manufacturing techniques can have a significant impact on the ultimate performance of a product. As indicated in Figure 3, the "property–mechanism–energy source–hybrid-AM process" (PMEH) approach presented by Webster S et al. gives direction for designing or researching various hybrid manufacturing processes [30].

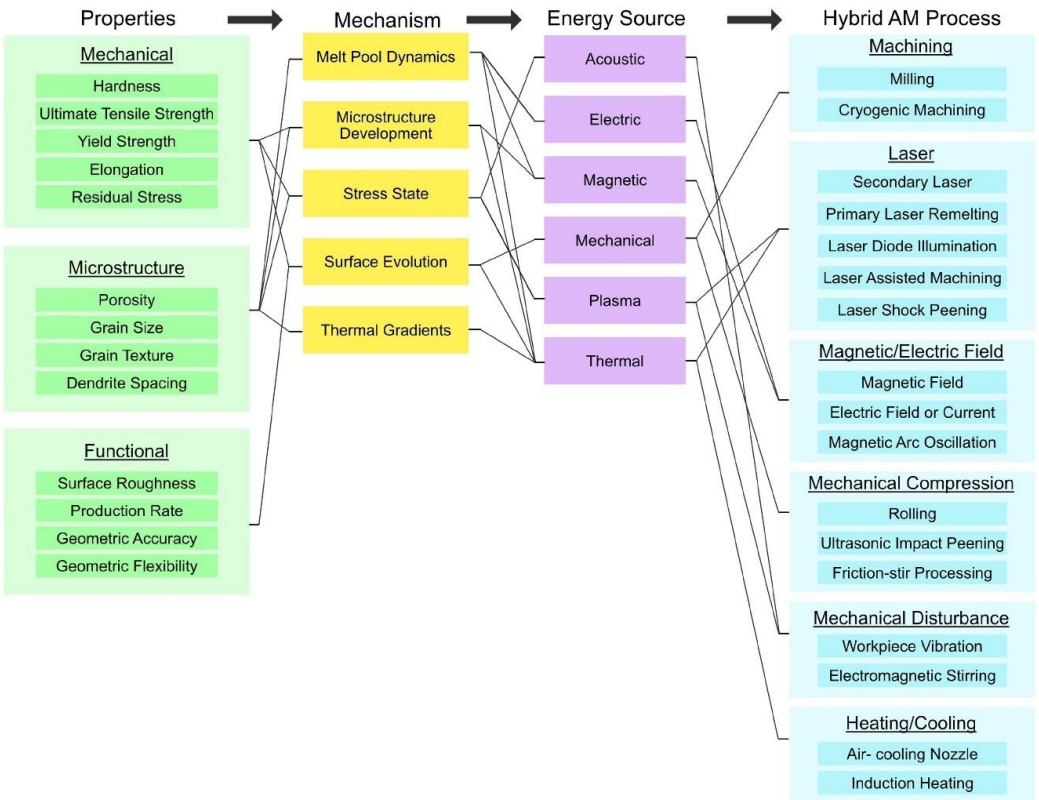


Figure 3. Property–mechanism–energy source–hybrid-AM process (PMEH) framework for hybrid AM classified by mechanism and energy source utilization [30].

The technologies of multi-process composite laser additive manufacturing, additive–subtractive composite manufacturing, multi-energy field composite manufacturing, and multi-material composite additive manufacturing are described in the following sections.

3.1. Multi-Process Hybrid Laser Additive Manufacturing Technology

3.1.1. Shot Peening

Laser additive manufacturing with shot peening provides shape- and property-controlling composite production techniques, which can alter the microscopic structures of metals, altering hardness, cracking, and mechanical properties. Shot peening is usually classified as laser shot peening (LSP), ultrasonic shot peening (USP), or mechanical shot peening (MSP).

LSP is one of the most essential post-treatment techniques for metallic materials. It is a technique that uses a high-powered laser beam to induce stress in materials in an effort to improve surface damage resistance, strength, and surface hardness [31–33]. The schematic is shown in Figure 4a. Dislocations occur and expand when the pressure of the laser shock wave surpasses the alloy's yield limit. Due to phase boundary delimitation and mechanical twin interactions in the dislocation walls, the microstructure is refined [34]. The creation of mechanical twins generated by LSP results in an increase in microhardness. With higher twin boundaries, the propensity for dislocation formation rises, leading to an increase in the tensile characteristics of LSP samples [35–37].

Chi J et al. [31] observed that the microhardness of DED-LSPed samples rose by 11%, 13%, and 12% in the top, middle, and bottom areas, respectively, which is intimately connected to the high density of dislocations in the grain structure. High-density dislocation lines may combine and create dislocation tangles (DTs) and dislocation walls (DWs). The existence of DTs and DWs implies dislocation sliding and accumulation during LSP, which results in the creation of sub-boundaries and the consequent refining of coarse grains (as shown in Figure 4b). Moreover, LSP introduces compressive residual stresses and plastic deformation into the machined surface, and the compressive stresses can offset some of the tensile stresses [38], reducing the crack extension rate and improving the tensile properties significantly [39] (as shown in Figure 4c, σt is the tensile residual stresses, and σc is the compressive stress state). The yield strength and ultimate tensile strength of the DED-LSPed samples increased to 1033 MPa and 1092 MPa (as shown in Figure 4d), respectively, which could be attributed to the increased work-hardening rate caused by dislocation and twin accumulation [35].

Kalentics N et al. [40] combined 3D laser impact peening with selective laser melting and discovered that integrated workpieces were transformed from tensile residual stresses to compressive residual stresses, which was extremely beneficial in improving fatigue life and reducing cracking phenomena. When compared to conventional LSP treatment, 3D LSP increased the size and depth of the compressive residual stresses significantly. Furthermore, due to the higher impact density on the machined surface, smaller spot sizes and higher overlap rates increase the magnitude and depth of the LSP-affected area when laser impact peening is performed (as shown in Figure 4e).

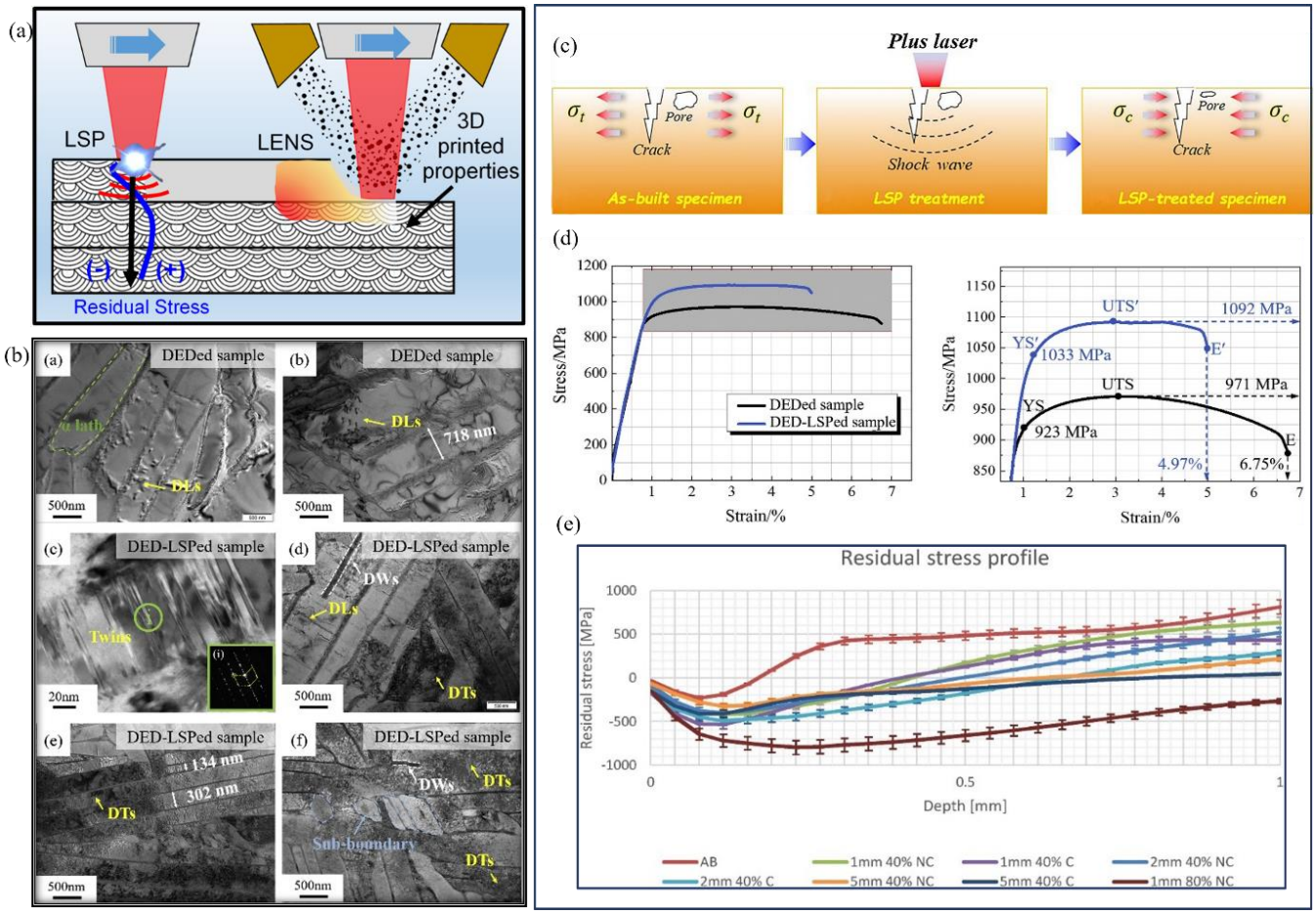


Figure 4. (**a**) Schematic diagram of AM with LSP hybrid technology [41]. (**b**) Microstructure comparison of DED and DED-LSP samples [35]. (**c**) Evolution of pores and residual stresses in samples under LSP treatment [42]. (**d**) Residual stress curves for 316L samples under LSP treatment. Spot size was 1 mm, and overlaps were 40% and 80% [38]. (**e**) DED and DED-SLP samples' stress–strain curves [35].

Ultrasonic shot peening (USP) employs ultrasonic vibration energy to propel the projectile at high speeds onto the surface to be processed. When compared to traditional peening technologies, USP offers the benefits of being non-polluting and allowing process controllability [43]. The schematic diagram of LAM with USP hybrid technology is shown in Figure 5a. The excellent corrosion resistance of USP-treated samples can be attributed to microstructure evolution, such as grain refinement and increased dislocation density, which results in the formation of a dense and homogeneous passive film. In general, smooth surfaces have better corrosion resistance relative to rough surfaces. At higher surface roughness, the compact and uniform passive film formed is more likely to break down, thus causing an increase in the corrosion rate [44]. Wu B et al. [45] investigated the effect of ultrasonic shot peening on the fatigue life of the titanium alloy Ti6Al2Zr1Mo1V (TA15). In the microstructure of USP-treated samples, a high density of dislocations and deformation twins appeared, restricting moving dislocations from reaching the free surface and delaying crack nucleation (as shown in Figure 5b), thereby reducing the rate of crack expansion.

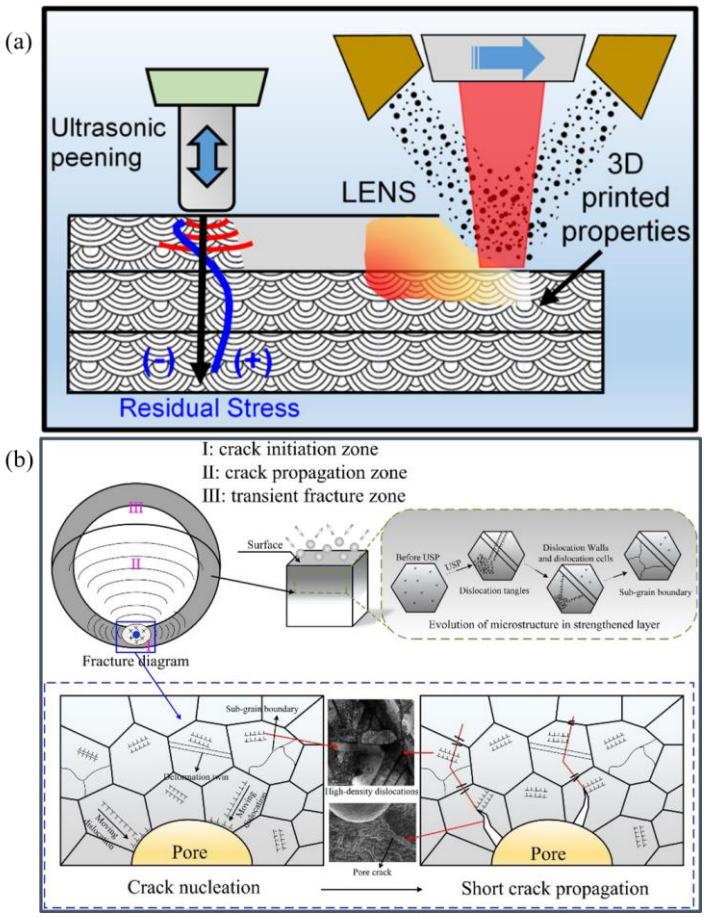


Figure 5. (**a**) Schematic diagram of AM with USP hybrid technology [41]; (**b**) effect of microstructure on crack extension [45].

Mechanical shot peering (MSP) is the application of a high-speed projectile blast to the surface of an object, causing plastic deformation and the creation of a hard layer with a certain thickness. The schematic diagram of AM with MSP hybrid technology is shown in Figure 6. The type of shot and the shot pressure used in mechanical peening have a detrimental impact on the microstructure and mechanical properties of additively manufactured components. Walczak M et al. [46] studied the effects of CrNi steel shot, nutshell granules, and ceramic beads as shot types and pressures of 0.3 and 0.6 MPa on component surface roughness, microstructure, hardness, wear, and corrosion. The results show that as the shot pressure increases, the surface roughness of the samples peened with ceramic beads decreases, the grain size is refined, and the corrosion resistance is improved. Uzan NE et al. [47] investigated the effect of shot peening on the fatigue resistance of specimens prepared for analysis by selective laser melting. The fatigue resistance properties of AlSi10Mg parts fabricated by selective laser melting (SLM) were found to be improved by optimal mechanical or electrolytic polishing of the shot-peened surface.

Based on a combination of laser, ultrasonic, and mechanical shot peening methods, coupled with laser additive technology, the particles are propelled at high speeds toward the target surface, causing the evolution of microstructures such as high-density dislocations to improve mechanical properties such as microhardness, residual stresses, and fatigue resistance. Laser shot peening processes can typically induce 4–5 times the compressive residual stresses, with greater depth, strength, and surface finish. The hybrid manufacturing technology based on shot peening is still in the continuous exploration stage, and the related grain refinement theory and process manufacturing mechanism require more in-depth research.

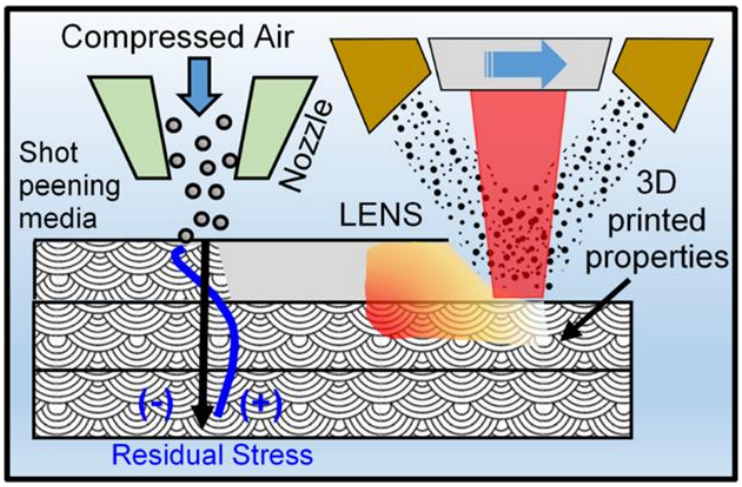


Figure 6. Schematic diagram of AM with MSP hybrid technology [41].

3.1.2. Laser Remelting

The laser remelting hybrid additive manufacturing process involves the deposition of metal powder onto the workpiece surface through laser-induced melting, followed by the localized heating of the material using the laser to facilitate the fusion of the molten metal powder with the substrate. Notably, during this process, the molten metal is influenced by Marangoni forces, leading to surface rearrangement and the promotion of a smoother texture. This methodology effectively upholds the advantageous features of laser additive manufacturing while simultaneously augmenting surface integrity and mechanical properties through remelting techniques [48]. The schematic diagram of AM with remelting hybrid technology is shown in Figure 7a.

Laser remelting melts coarse columnar grains, inhibits grain growth, and results in a more uniform composition distribution, optimizing the alloy's microstructure, reducing surface roughness, and improving mechanical properties such as wear and corrosion resistance [49–51]. In general, there are two types of laser remelting treatments, surface remelting and interlayer remelting, with surface remelting applied only to the last layer of the part. Interlayer remelting allows for the rescanning of other layers in addition to the surface layer.

The thermal history and local temperature gradients of the melt pool are significantly influenced by various laser remelting process parameters (laser power, scan spacing, scan cycles, scan speed) [52]. The LR process can reduce intra- and interlayer porosity at appropriate energy densities [53,54]. Yu Z et al. [55] investigated the differences in the mechanical properties of microstructures at different energy densities (8.7, 15.0, and 21.6 J/mm^2). The samples had the lowest porosity and the best elongation anisotropy at a laser energy density of 15 J/mm^2. Xin B et al. [56] studied the microstructure and mechanical properties of various laser powers. While the decrease in porosity improves cohesion and suppresses deformation within the clad layer, the progressive increase in laser power allows residual porosity to escape. At 1200 W, the sample has the lowest porosity and relatively homogeneous grain size (as shown in Figure 7b), which accounts for the higher average hardness, ultimate tensile, and yield strengths after remelting. Furthermore, the LR treatment facilitated the uniform dispersion of the alloying components, hence guaranteeing adequate wetting and diffusion time for the melt pool, resulting in specimens with a smoother surface [52]. As demonstrated in Figure 7c, the Sa values on the top and sides of the samples were lowered to 8.9 μm and 4.5 μm, respectively.

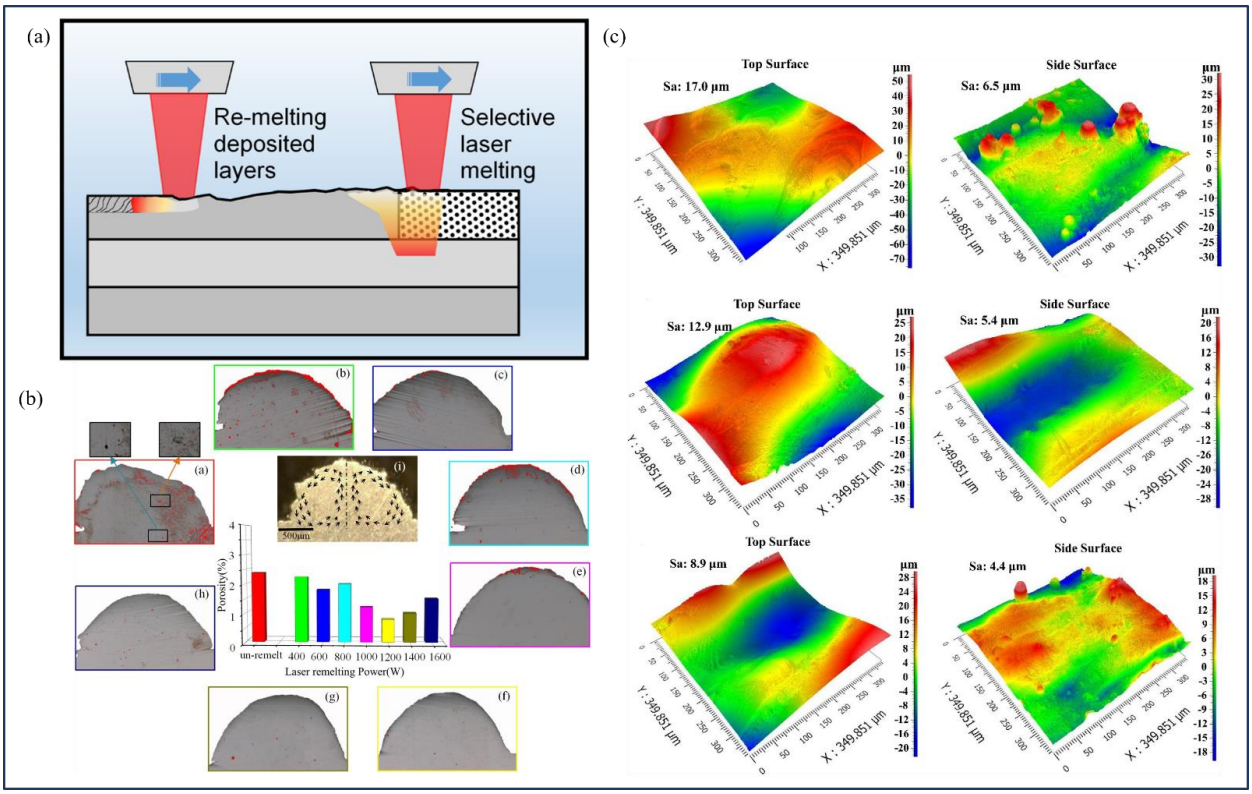


Figure 7. (**a**) Schematic diagram of AM with remelting hybrid technology [41]; (**b**) effect of microstructure on crack extension [56]; (**c**) variation in surface roughness Sa of different samples [52].

When laser remelting technology is utilized for surface modification, it typically entails the localized heating and rapid cooling of the material using a high-energy laser beam. This process facilitates the melting and re-solidification of the surface, leading to defect elimination and enhanced surface finish, as well as the improved ductility and corrosion resistance of the material. It is important to note that laser remelting may induce changes in the material's microstructure and chemical composition, thereby potentially impacting its chemical properties and overall performance. Laser remelting changes the surface's elemental composition by increasing the concentration of aluminum and vanadium on the surface, reducing the corrosion resistance and biocompatibility of the Ti6Al4V alloy [57]. This hybrid manufacturing technology has the potential to produce alloys with medium to high entropy and excellent properties [58,59]. Cai Z et al. [59] attempted to create a high-entropy Ni-Cr-Co-Ti-V alloy coating. The high coefficient of friction and low wear mass loss of the remelted high-entropy alloy coating were discovered to improve wear resistance and could be used in the wear protection of brake pad coatings.

3.1.3. Forging

Two prospective production methods for hybrid forging–additive manufacturing technologies have been developed: The first is the deposition of complicated structural components onto a forged substrate using laser additive manufacturing technologies. In contrast to conventional additive manufacturing techniques, the planned forged substrate is kept as a component of the finished item. The second utilizes forging technology to treat the deposited components following additive manufacturing [60].

This hybrid manufacturing approach is based on the deposition of complicated components on forged substrates. This production technique makes full use of the high performance and flexibility of LAM as well as the efficiency and low cost of forging traditional components. By combining forging and deposition methods, it is possible to acquire more advantageous mechanical qualities, allowing for the economical production of vast and

complicated components [61]. The microstructure of the HAZ between the substrate and deposited area is crucial to the performance of the final component [62]. Ma J et al. [61] found that the microstructure of the HAZ area of hybrid-fabricated Ti6Al4V samples was significantly reliant on the intricate thermal history of the deposition process (peak temperatures at different locations and at different cooling rates). Bimodal microstructures, including equiaxed and lamellar, appear when the bottom HAZ is distant from the melt pool and the temperature is below the phase transition temperature; with subsequent heating cycles, the secondary phase precipitates in the phase, and a ghost structure appears at the top HAZ as a result of alloying element diffusion (as shown in Figure 8a,b). The ghost structure has a greater microhardness than the lamellar phase and may enhance the yield strength, hence improving the mechanical characteristics of the products. The microhardness profile of Ti6Al4V alloy fabricated by hybrid AM with forging is shown in Figure 8c [61].

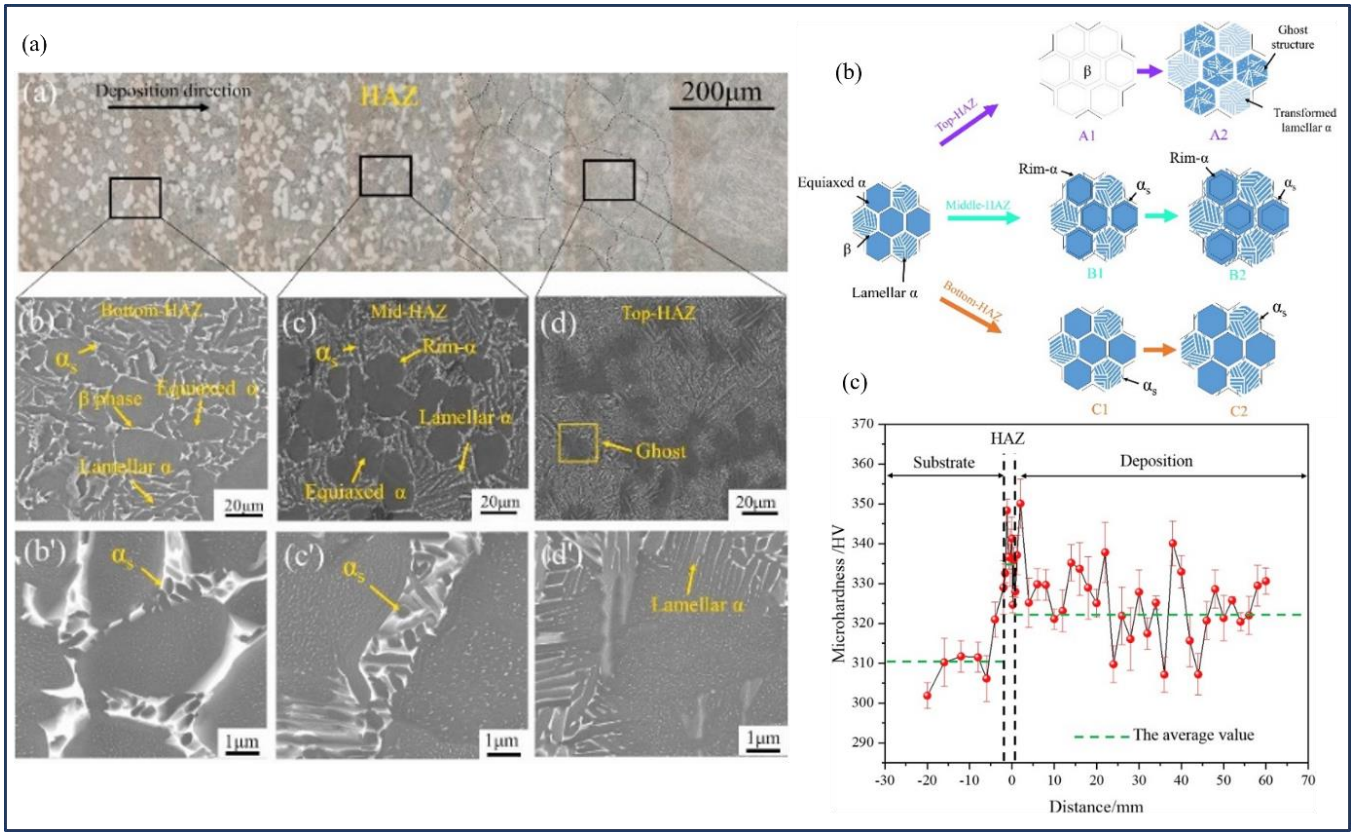


Figure 8. (**a**,**b**) HAZ microstructure and evolutionary pattern; (**c**) microhardness profile of forged Ti6Al4V alloy [61]. (a) HAZ(heat affected zone) microstructure and evolutionary pattern (b) bottom-HAZ, (c) middle-HAZ, (d) top-HAZ. (b'–d') is a localized enlargement of b–d.

The laser forging hybrid process is essentially the simultaneous processing of metal parts with two functionally different laser beams coordinated with each other, one laser beam performing additive manufacturing and the other laser beam acting on the deposition zone, and forging using the shock wave generated by the pulsed laser; both processes are carried out simultaneously [63]. The porosity of the deposited layer is eliminated by laser forging, resulting in improved quality and properties [64,65].

Wu, D. et al. [66] applied an independently developed synchronous hammer-forging-assisted laser-directed energy deposition device (as shown in Figure 9a). In Figure 9b, the C and D layers that have undergone simultaneous hammer forging yield tinier grains with lower-angle grain boundaries than the untreated A and B layers. The microstructure and mechanical properties of the laser additive manufactured prototypes were modulated due

to the grain refinement and work-hardening effect produced by low-pressure hammer forging, verifying the suitability of this technique for weakly rigid components. The manufacturing samples are shown in Figure 9c. Wang Y. et al. [67] evaluated the mechanical properties of Ti6Al4V titanium alloy specimens prepared through the additive manufacturing and forging of various power preparations. Tensile properties and microhardness were found to vary slightly in four regions of the microstructure morphology, with the best tensile properties at a laser power of 1000 W.

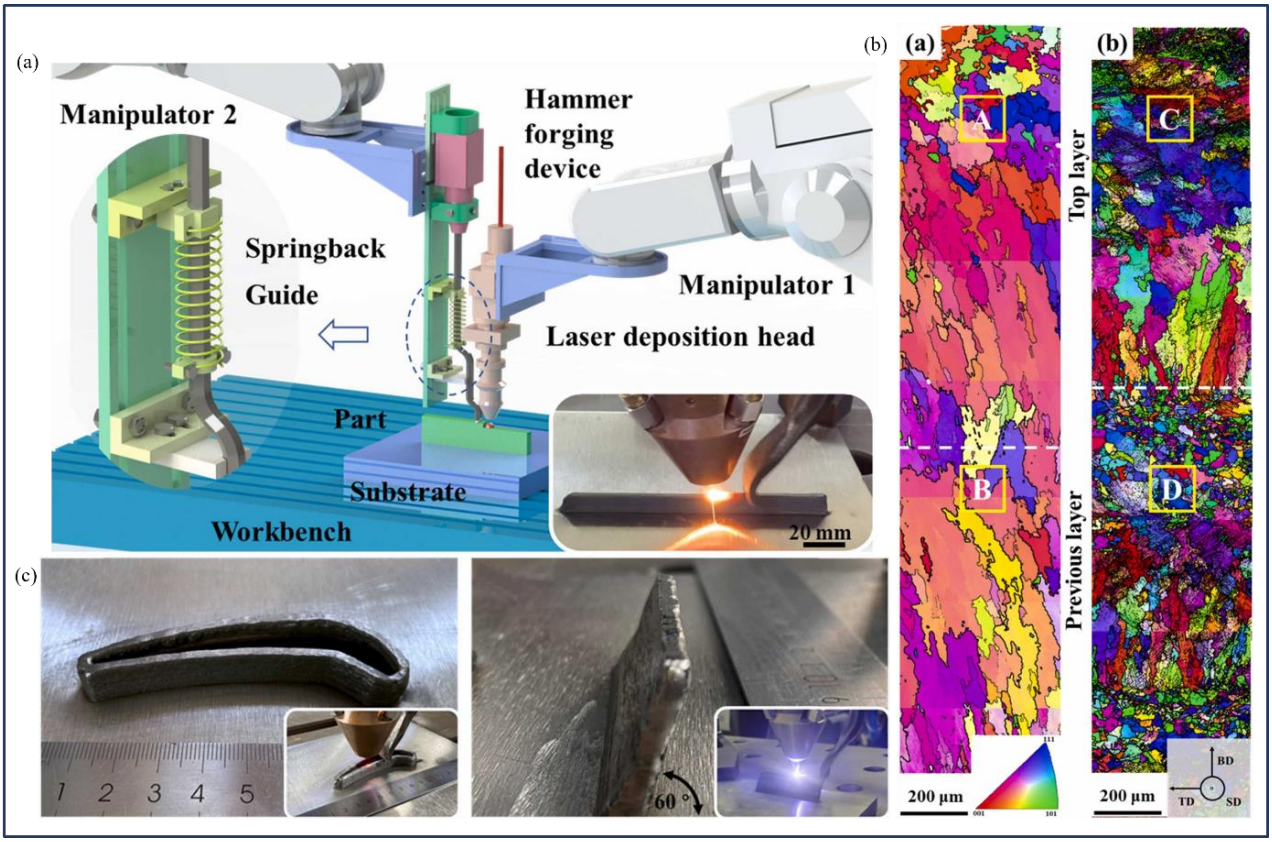


Figure 9. (**a**) LAM with forging device; (**b**) EBSD diagrams of structural evolution; (**c**) LAM with forging manufacturing samples [66]. (**a**) the top two layers of LDED sample; (**b**) is a synchronous-hammer-forging-assisted laser directed energy deposition (SHLDED) layer (A) is the top layer of LDED sample, layer (B) is the previous layer of LDED sample, layer (C) is the top layer of SHLDED sample, and layer (D) is the previous layer of SHLDED sample.

3.1.4. Arc Additive Manufacturing

Laser–arc hybrid manufacturing is a novel processing method that combines laser and arc heat sources. Figure 10a shows the processing principle. The interaction principle of the laser and arc is that arc heating improves the laser utilization rate, and the laser's introduction plays a role in stabilizing and inducing the arc [68]. This hybrid processing technology is commonly used in shipbuilding, transportation, and pipeline processing. Gong M et al. [69] conceived an additive laser oscillation arc manufacturing method. The prepared samples' surface roughness was reduced to 20% of that of arc additive manufacturing, and porosity was effectively suppressed. Furthermore, uniform tensile properties with minimal tensile anisotropy were obtained. The deposited surface morphology and sidewall surface accuracy are shown in Figure 10b.

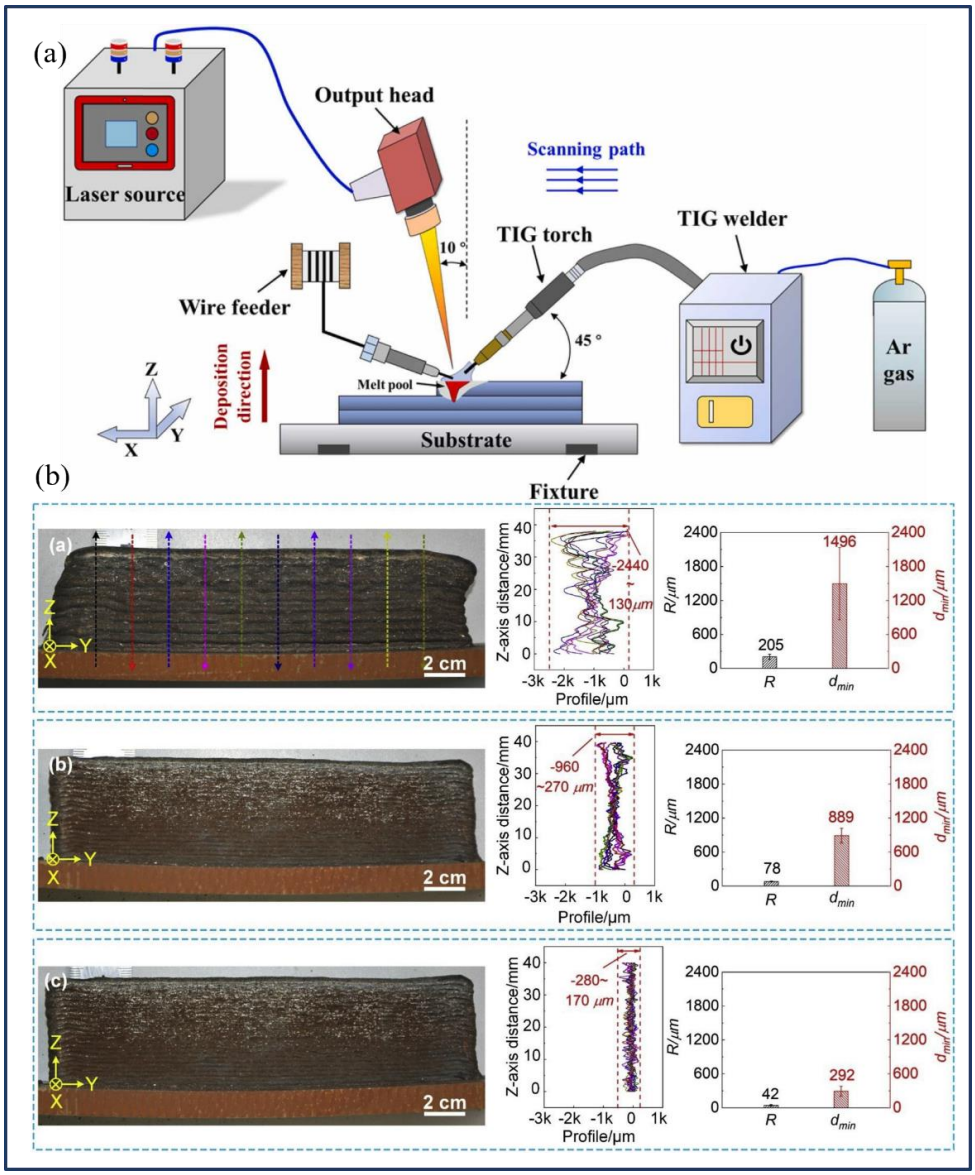


Figure 10. (**a**) Laser–arc hybrid additive manufacturing schematic diagram [68]; (**b**) surface accuracy of the deposited surface profile and sidewalls [69]. (a) wire-arc additive manufacturing (WAAM), (b) laser-arc hybrid additive manufacturing (LHAM), (c) oscillating laser-arc hybrid additive manufacturing (O-LHAM).

3.2. Additive–Subtractive Hybrid Manufacturing

Additive–subtractive hybrid manufacturing technology forms three-dimensional solids through successive steps of material accumulation and removal. Layer-by-layer deposition is a common additive manufacturing technique that can produce complex-shaped parts without limitations but with poor surface integrity. In terms of part formation quality, subtractive machining performs admirably. Combining the two processing technologies and fully utilizing their advantages can greatly improve efficiency and realize reasonably accurate part preparation. Laser additive and subtractive hybrid manufacturing technology also provides an effective repair measure for the remanufacturing of complex parts, such as damaged blades [70].

Hybrid forms of additive and subtractive materials can be classified as separate or collaborative. Separate forms involve cutting the part after it has completed the additive manufacturing process. The dimensional tolerances and surface integrity of additive parts are adversely impacted by the heat accumulation effect of the additive manufacturing

process. Subtractive machining, on the other hand, is conducted in a cold state with high machining precision [71]. Synergetic manufacturing, which alternates between additive and subtractive processes, can effectively reduce dimensional deficiencies accumulated during the additive process and improve manufacturing accuracy. However, subtractive machining is performed in a hot state, which eventually causes tool wear [72].

In PBF-based additive and subtractive hybrid manufacturing technology, a metal powder is melted by laser powder bed additive manufacturing technology and the sintered surface is machined with a milling tool to make it smooth after accumulating several layers one by one. Subsequently, the milling and cutting processes alternate until the process takes shape. This technique effectively eliminates geometric defects in the additive manufacturing process and significantly improves the shaping quality of the machined part [73].

Sodick [74] has developed a hybrid additive and subtractive manufacturing machine, OPM 250L(Sodick, Kanagawa, Japan), which has significant advantages in the design and manufacture of injection molds and can produce parts with complex geometries. The layout and function of cooling channels have been improved in processing to prevent thermal build-up in specific areas and to significantly reduce the molding cycle time.

Matsuura [75] has created the LUMEX-Avance 25 (Matsuura, Fukui, Japan), a hybrid production machine that combines PBF and milling machining technologies. Increased powder supply speeds, as well as optimized control software and machining pathways, allow for faster processing. Furthermore, the system can fully automate powder delivery, collection, and reuse. Based on the machining qualities of the LUMEX series machines, they are frequently employed to produce highly functional molds. In addition, lightweight elements, such as interior cavities or webs, may also be easily built. Figure 11 shows LUMEX-Avance 25-machined parts [76].

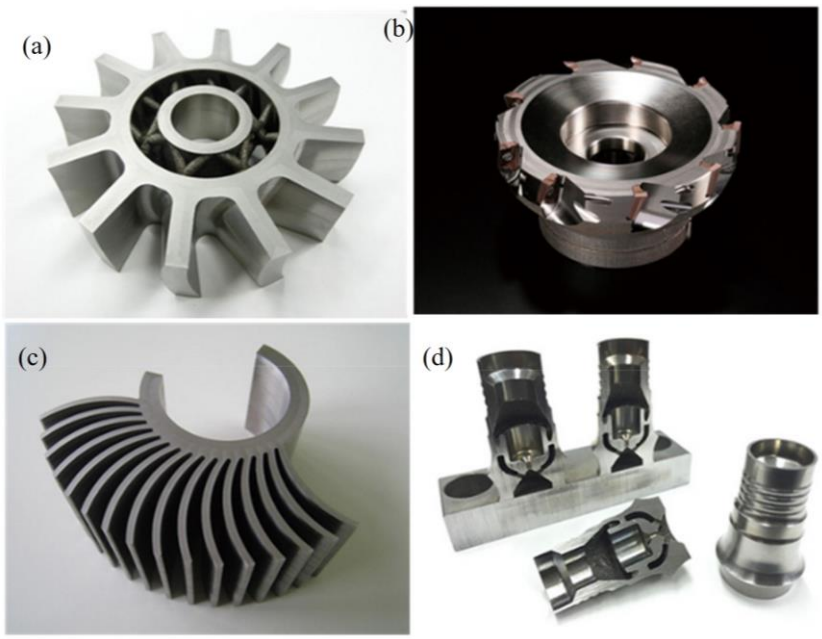


Figure 11. LUMEX-Avance 25-machined parts [76]. (**a**) Blisk; (**b**) Face Mill; (**c**) Cooling Fan; (**d**) Jet Engine Nozzle.

By combining LDED technology with CNC machining platforms, DED-based hybrid machining technology for additive and subtractive materials is created. As indicated in Table 1, this sort of equipment provides outstanding process flexibility to fulfill a wide range of process production needs.

Mazak [77] has deployed two types of machining heads. The first is a precision LDED machining head capable of machining complex shapes of parts with high precision, and the

second is a high-speed LDED machining head capable of high speed and efficient fusion cladding. The LDED machining head includes a standard tool magazine for subsequent additive material manufacturing. The device provides excellent process flexibility to satisfy a broad range of process production demands. The Mazak INTEGREX i-400 AM (Mazak, Aichi-Pref, Japan) machine is frequently used for surface coating and component repair.

DMG MORI [78] has developed a LASERTEC 125 machine (DMG MORI, Bielefeld, Germany), which can transmit a wide range of powder materials, such as bronze, copper alloys, high-speed steel, and nickel-based alloys, for the fabrication of gradient functional structural parts, providing strong support for tailoring the parts' individual functional structures. Figure 12 depicts a fabricated cutter, valve body, and multi-material heat exchanger.



Figure 12. LASERTEC 125-machined parts [78]. (**a**) Cutter, (**b**) valve body, (**c**) multi-material heat exchanger.

3.3. Multi-Energy Field Hybrid Manufacturing Technology

Multi-energy field hybrid additive manufacturing technology applies ultrasound, electromagnetic, and laser energy sources to the additive manufacturing process. The microstructure and macroscopic properties of the deposited parts are significantly improved [79]. The more widely used ones are ultrasonic vibration and electromagnetic.

3.3.1. Ultrasonic Vibration-Assisted Additive Manufacturing

The introduction of ultrasonic vibrations into laser additive manufacturing techniques can dramatically improve melt pool flow characteristics [80]. As indicated in the schematic design in Figure 13a, ultrasonic vibration has two essential mechanisms of action: acoustic flow and cavitation; the principle of action and improved performance is shown in Figure 13b. Acoustic flow is the constant flow caused by the attenuation of ultrasonic waves, which facilitates heat exchange and material movement, aids in the reduction in temperature gradients within the melt pool, and balances the elemental distribution [81]. Cavitation is a constantly changing process. When an oscillating, expanding compression bubble reaches a resonance size dictated by the frequency of the sound field, it develops rapidly as it efficiently absorbs heat within the melt pool. The bubble expands, and the stress on its liquid surface combines with the next compression wave to cause it to explode in a sub-microsecond time frame. Inside the melt pool, temperatures of 5000 K and pressures of about 1700 atm are created [82]. The creation of high-pressure shock waves and high-speed micro-jets can break the forming dendrites and scatter them uniformly in the melt pool, resulting in fine and uniform solidification organization. Furthermore, the formation of cavitation bubbles absorbs a huge quantity of heat at the same time, leading to the local undercooling of the melt and an increase in local subcooling. Changing the degree of subcooling and the nucleation rate during solidification affects the crystallization of molten metal [83,84].

This approach may eliminate deficiencies induced by fast solidification, refine microstructure grains, and enhance the mechanical characteristics of additively made components, such as fatigue, corrosion, and tensile properties [85,86]. Table 1 summarizes the frequently employed powders, as well as the improvements in microstructure and mechanical properties induced by ultrasonic treatments.

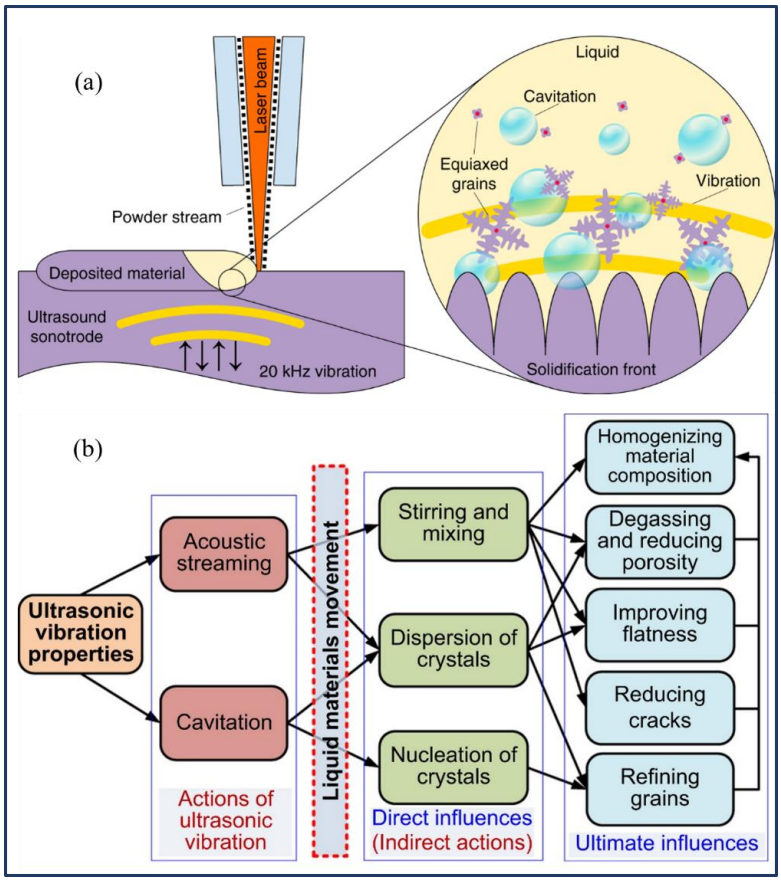


Figure 13. (**a**) Schematic diagram of ultrasonic vibration compound laser additive manufacturing [87]; (**b**) action and influences of ultrasonic vibration in UV-A [88].

The morphology and grain size of the microstructure are closely related to the temperature gradient (G) and the solidification rate (R), according to solidification theory; the higher the G × R value, the finer the microstructure morphology, and the lower the G/R value, the better the formation of equiaxed dendrites [89]. The grain number density, which is closely connected to the nucleation density, is frequently used to assess the influence of ultrasound on grain refinement [90]. Zhu L et al. [85] theoretically analyzed the effect of ultrasound on the degree of subcooling and transient high pressure as vibrational parameters (amplitude and frequency) increased. When ultrasonic treatment is used, increasing the degree of subcooling increases the nucleation rate, resulting in the production of more grains. This benefits grain refinement. The average grain size is 0.522 times the non-vibrating grain size when the amplitude is 25 μm.

The ultrasonic vibration process parameters have a substantial influence on the enhancement of the microstructure and mechanical characteristics of laser cladding components. The degree of subcooling is affected by the frequency and amplitude of the ultrasonic waves, and an increase in the degree of subcooling leads to an increase in the nucleation rate. The frequency magnitude influences both the magnitude and the period of the sound pressure, whereas the amplitude simply affects the magnitude of the sound pressure [85]. The intensity of ultrasonic cavitation and the velocity of ultrasonic acoustic flow will steadily grow as the ultrasonic power increases, increasing the pace at which the high-temperature fluid travels toward the bottom of the melt pool [89].

Acoustic flow and cavitation's dynamic behavior causes substantial transient pressure and temperature variations inside the liquid medium. The ensuing ultrasonic vibrations improve Marangoni convection inside the melt pool, encouraging better stirring and mixing of the components within the melt pool and increasing the melt pool size [88]. The coating's dilution ratio rises somewhat with increased ultrasonic power, allowing for

stronger interfacial adhesion between the component and the substrate [89]. According to Li M et al. [91], the ultrasonic cavitation intensity is low at 600 and 700 W ultrasonic power, and the acoustic flow range is limited, with the WC particles concentrated toward the bottom of the cladding due to gravity. The application of 800 W of extra ultrasonic power leads to a more uniform dispersion of the WC particles, which improves the precipitated carbides and results in high hardness and wear resistance of the cladding at this point. Higher ultrasonic frequencies cause more ultrasound in the melt pool during solidification, which contributes to a better distribution of acoustic pressure and, as a result, a reduction in grain size [92]. It has been shown that the triggering of cavitation by ultrasonic vibrations generally has a critical value (cavitation threshold). Above a particular amount, the threshold rises, reducing the degree of cavitation and thus increasing the number of entrained bubbles and hence the porosity. This is one of the reasons why, as the ultrasonic frequency increases from 33 kHz to 41 kHz, the wear rate of IN718 components increases [93].

Ultrasonic vibrations have been proven to manage residual stresses. The residual stress in the clad layer is constantly changing at different ultrasonic powers and is cut to a bare minimum at 600 W [93]. Internal microstructure flaws generated by air bubbles entrained during the manufacturing process may influence residual tension. According to Zhang, D et al. [94], ultrasound facilitates air bubble escape, resulting in less penetration of the created micro-pores to lessen residual stresses.

Table 1. Summary of ultrasonic vibration-assisted additive manufacturing system.

Materials (Substate/Cladding Layer)	Additive Manufacturing Processes	Optimal Process Parameters for Ultrasonic Vibration	Microstructure Evolution	Performance Improvements	Refs.
45 steel/ Ni-WC ceramic hybrid coatings;	Laser cladding	800 W	Refinement of dispersed dendrites	Improved surface hardness and wear resistance	[91]
Ti6Al4V/Ti + B4C	Laser cladding	400 W	Dendrite gradually becomes granular	The friction performance is improved, the friction coefficient is 50% lower, and the wear resistance is 1.2 times that without ultrasonic vibration	[95]
5CrNiMo steel/FeTi30 + FeCr70	Laser cladding	300 W	The ceramic particles are more evenly distributed, and the dendrite size is refined	Reduced surface roughness, increased coating dilution, improved hardness, and increased high-temperature oxidation resistance	[89]
45 steel/ Inconel 718	Laser cladding	A = 25 μm	The average grain size is 0.522 times that of the non-vibration grain size.	Effectively reduced the porosity, decreased the friction coefficient to 0.628 times that without ultrasonic waves, and improved the microhardness and wear resistance	[85]
Low-carbon steel/In 718	Laser-engineered net shaping	25 kHz	The grain size is smaller, and the Laves phase changes from a long columnar to a granular shape	Excellent interfacial bonding ability, high microhardness, low porosity	[93]

Table 1. *Cont.*

Materials (Substate/Cladding Layer)	Additive Manufacturing Processes	Optimal Process Parameters for Ultrasonic Vibration	Microstructure Evolution	Performance Improvements	Refs.
WTaNbMo refractory high-entropy alloy coating/ IN718	Laser cladding	20.5 kHz; A = 9.6 μm	The average grain size of the coating decreased from 0.48 μm^2 to 0.30 μm^2; microstructure and element distribution are more uniform	Reduced residual stress in the coating, increased hardness, and improved high-temperature friction properties	[96]
Ti substrate/NiTi	DED	25 kHz; 40 W	Homogenization of microstructure, reduction in secondary phases, and refinement of grains	Microhardness and Young's modulus enhancement	[94]
Ti6Al4V/In 625	DED	20 kHz; 30 μm	Transition from coarse columnar crystals to fine equiaxed crystals (100 μm), microstructural homogeneity	Yield stress and tensile strength increased by 12%	[87]
Low-carbon steel/AISI 630 stainless steel	Laser-engineered net shaping	41 kHz; 60 W	Reduction in porosity and microcracking	Increased tensile properties and hardness, increased powder utilization, and improved bonding of the deposited part to the substrate	[88]

3.3.2. Electromagnetic-Assisted Laser Additive Manufacturing

Electromagnetic-assisted laser additive manufacturing involves the utilization of electromagnetic stirring technology in the laser additive manufacturing process. The flow, heat transfer, mass transfer, nucleation, and growth of crystallization of the metal melt are all affected by the electromagnetic force formed by the induced current and magnetic field.

Zhou H et al. [86] investigated and compared the effects of static and alternating magnetic fields on the microstructure, weave, and mechanical properties of SS316L. Under static magnetic fields, the crystal texture of cellular dendrites along the building direction is suppressed. The alternating magnetic field accelerates thermal diffusion and increases the cooling rate, allowing the grain size and spacing to be reduced. An appropriate alternating magnetic field can also be applied to achieve a specific crystalline phase orientation along the solidification direction of the additively manufactured metal part.

Furthermore, in the absence of a magnetic field, Marangoni convection promotes bubble formation and traps bubbles in vortices. The fast, unstable Marangoni will be suppressed by the damping effect of a static magnetic field, and the bubbles trapped in the vortex will be driven to the melt surface and released into the air [97,98]. This is shown in Figure 14. The damping effect of magnetic energy on the Marangoni causes grain refinement, lowered cellular dendrite spacing, and significantly reduced porosity at the deposition site, resulting in improved strength and ductility [99,100].

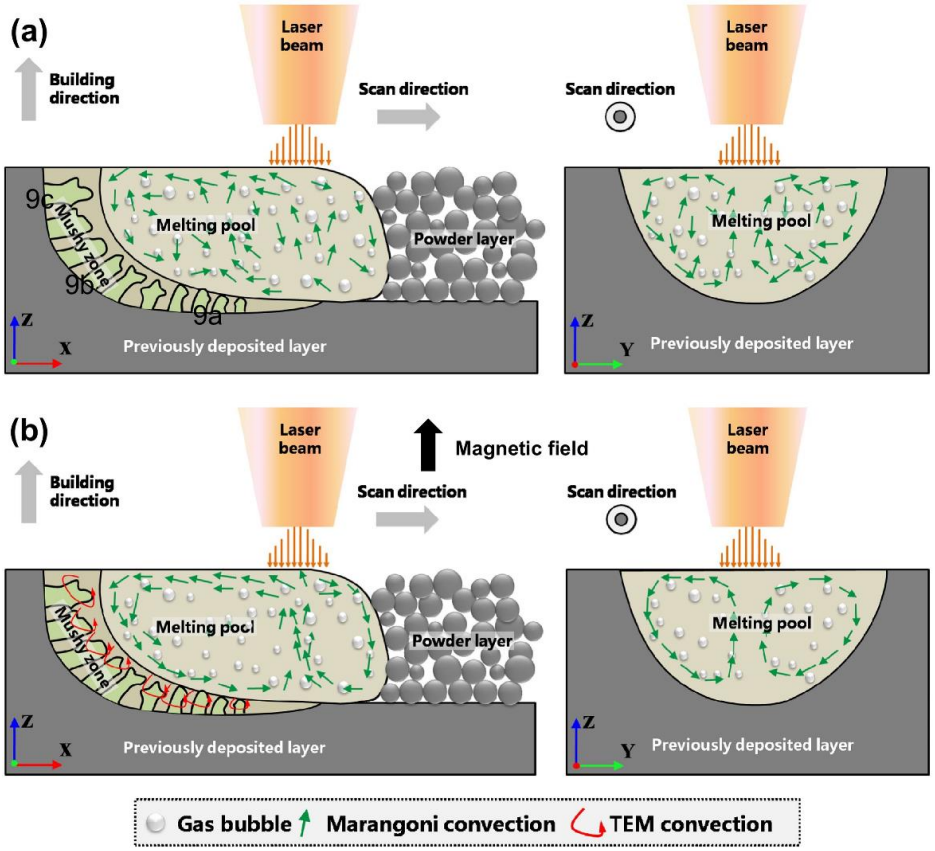


Figure 14. Diagram of the effect of the magnetic field on the flow of a liquid [98]. (**a**) the Marangoni convection without magnetic field; (**b**) the Marangoni convection with the application of a static magnetic field.

3.4. Multi-Material Hybrid Additive Manufacturing Technology

Laser additive manufacturing technology uses layer-by-layer deposition to produce components, which brings new application prospects for the structural design of multi-material components [7]. Not only are manufacturing costs reduced, but different powder materials can be deposited at specific locations to improve part performance variation [101]. Multi-material additive manufacturing technology currently has significant application potential in aerospace, automobile, medicine, energy, and optoelectronics [102,103].

Material transport in a multi-material preparation process necessitates not only a consistent powder flow rate but also a powder ratio that can be selectively adjusted in real time. As a result, developing powder conveying systems for various mixing ratios is critical. Zhang X et al. [104] developed a system that allows two powders to be mixed and dispensed in a controlled ratio to study Cu10Sn copper alloy powders and soda–lime glass powders. The system dispenses the mixed powder using an ultrasonic vibratory feeding system to maintain a constant powder flow rate. The two upper vibratory feeding systems deliver the two types of powders into the lower mixing chamber. The amount of powder dispensed is controlled by the dispensing time based on the constant flow rate. In addition, a double-hopper system was used to achieve the preparation of a multi-material structural part. These two hoppers inject the required amount of powder into the mixing chamber by means of a piezoelectric sensor. The measured powder is mixed by rotating vanes. The mixed powder is discharged into the lower hopper, where the piezoelectric sensor is calibrated to release the desired amount by controlling the vibration time.

Bimetallic structures, gradient path transitions, and the introduction of intermediates are the most common fabrication methods used in the design of structures for multi-material additive manufacturing [5,105]. The following figures depict the various fabrication methods and prototypes produced. Figure 15a depicts the direct combination of two metals.

Heer, B. et al. [106] achieved graded magnetic functionality by transitioning from non-magnetic austenitic stainless steel 316 to ferritic stainless steel 430. However, in bimetallic structures, differences in the thermal physical parameters of dissimilar metals (e.g., thermal expansion/shrinkage coefficients, lattice mismatch) frequently affect the quality of interfacial bonding, causing interlayer porosity, delamination, and cracking in the part and deteriorating mechanical properties [107]. Gradient path transition and the strategy of introducing intermediates can significantly mitigate these phenomena.

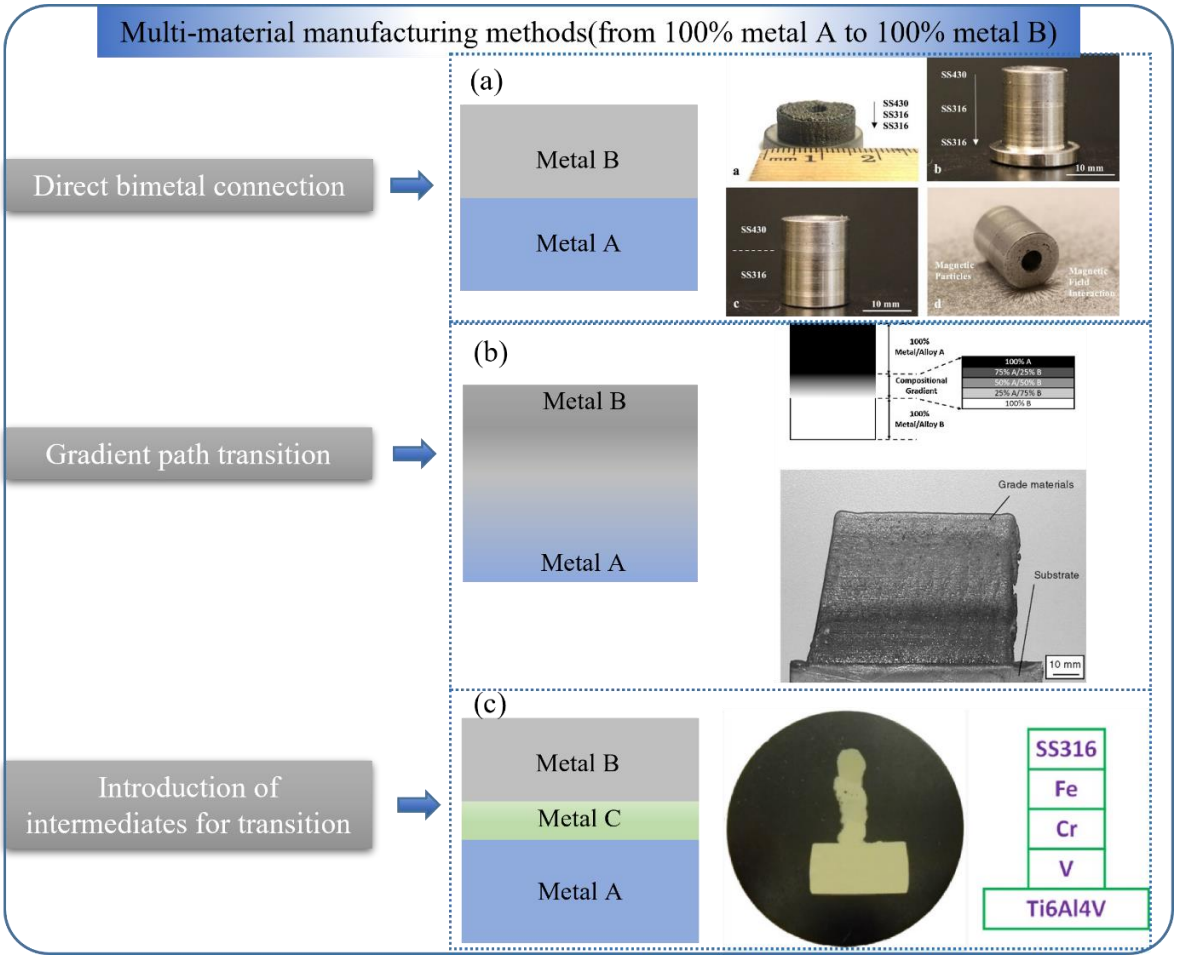


Figure 15. Common multi-material strategies used by LAM systems: (**a**) direct bimetal connection method and manufacturing samples [106]; (**b**) gradient path transition method and manufacturing samples [5]; (**c**) introduction of intermediates for transition and manufacturing samples [108].

Figure 15b shows a multi-material functional gradient manufacturing (MFGM) structure with a certain gradient transition ratio from 100% metal A to 100% metal B. MFGM is defined as the continuous modulation of a material's chemical composition, microstructure, physical characteristics, and other factors along its thickness or length direction, resulting in a spatial gradient of its physical and chemical properties according to design requirements [7]. The functional gradient structure can considerably alter the stress distribution around the fracture, influence the size of the local plasticizing area during fatigue crack propagation, and improve the uniformity of the residual stress distribution [109,110].

Nonetheless, the production of brittle intermetallic compounds along the component gradient of FGM still leads to part failure. For example, the direct connection of titanium alloy with stainless steel frequently results in the development of Fe-Ti intermetallic compounds (e.g., FeTi and Fe_2Ti), generating fissures at the interface [111,112]. As a result, the effective prediction of the metal phases that may be created by the multiple materials used

during the manufacturing process, in order to avoid detrimental brittle metal phases and optimize the gradient material path, is the key to improving the interface's metallurgical bonding capacity [113]. On the one hand, isothermal multicomponent phase diagrams and CALPHAD computations of phase equilibria can provide accurate forecasts of phase formation [114]. The Gibbs energies of all stable and metastable phases are created as parametric functions and evaluated using experimental and theoretical data acquired from first-principles calculations using generalized density functional theory [115].

Bobbio LD et al. [116] used CALPHAD to calculate the phase equilibria for the Fe-Cr-V and Fe-Ti-V systems, as shown in Figure 16a,b. Except for a minor percentage of the C14 phase predicted in the transition from 25 vol% Ti-6Al-4V/75 vol% V to 25 vol% SS304L/75 vol% V, the predictions are in good agreement with the calculated predictions of the experimentally determined phases. To avoid the formation of hard and brittle intermetallic compounds in the Fe-Cr-V system's phase-substable state, Reichardt et al. proposed two effective circumvention paths: using discontinuous gradient transition components to leap over this range and bypassing the phase field by adding a fourth component (e.g., pure Cr) [117], as shown in Figure 16c. This paper offers positive guiding suggestions for optimizing gradient pathways. A Scheil Ternary Projection (STEP) diagram is another non-equilibrium phase diagram that takes into account the considerable expansion phenomenon during melt melting under rapid cooling conditions and is used to create the ideal composition gradient to prevent brittle metal phases, as shown in Figure 16d. In STEP diagrams, the phase field simply reflects the presence of the phase, not its composition, which might vary depending on its position in the total solidified material position, not its phase fraction [118].

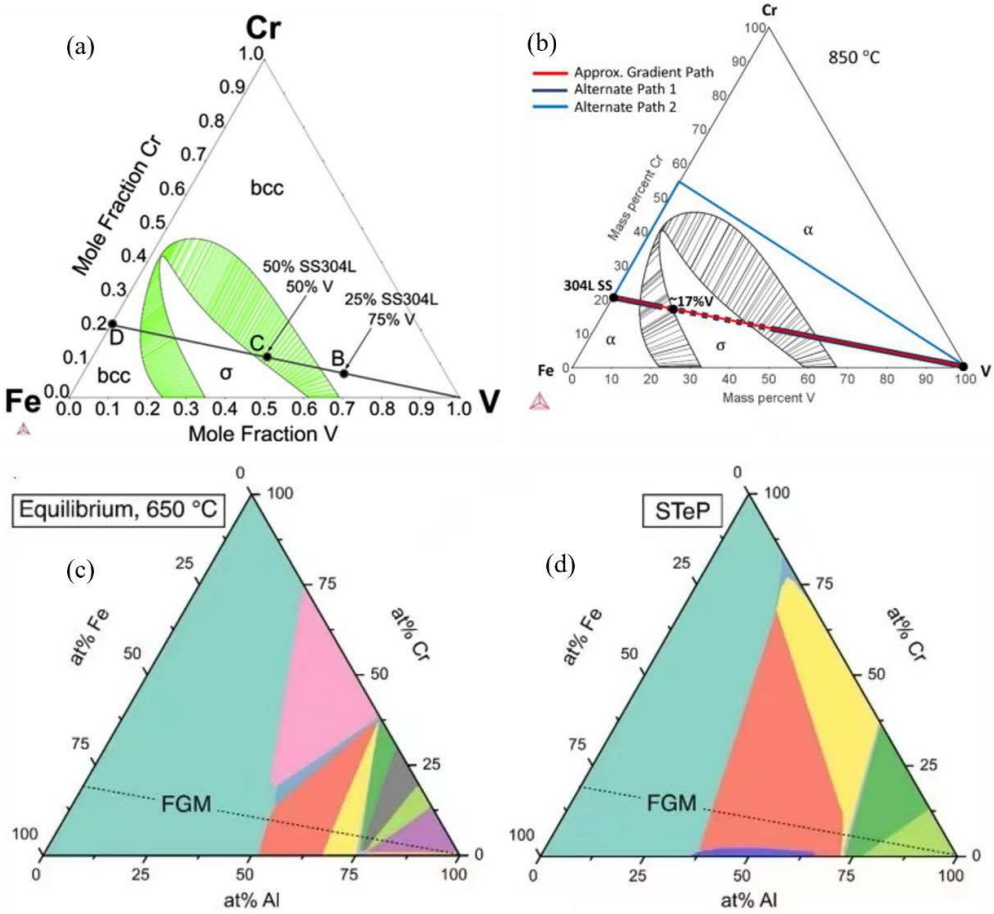


Figure 16. Multi-material gradient path optimization method. (**a**) Ternary phase diagram for Fe-Cr-V system prediction [116]; (**b**) different paths to circumvent brittle metal phases [117]; (**c**) equilibrium Fe-Cr-Al isothermal ternary diagram at 650 °C; (**d**) Fe-Cr-Al STeP diagram [118].

Figure 15c depicts a multilateral structural design by using intermediates, primarily single-element particles or multiple elements. Martin JH et al. [119] discovered that adding nanoparticles can induce non-homogeneous nucleation and accelerate the development of equiaxed grains. This has an effect on the liquid metal's solidification process, resulting in fewer massive columnar crystals and intermittent fissures. Moreover, Li, L. et al. [120] used laser fusion deposition to create Ti6Al4V composites reinforced with TiC particles (TiCp). The high hardness of functionally graded composites may result in superior wear resistance.

The critical question in the deposition of multi-material functional gradient structures is how to achieve excellent interfacial bonding. Poor interfacial bonding can cause porosity, delamination, and other deficiencies between the layers of the part, significantly reducing its mechanical properties. Differences in the thermal expansion/shrinkage and lattice matching of dissimilar materials can have an impact on the surface integrity of the part [121]. Brittle intermetallic compounds can form along the FGM component gradient, deteriorating the quality of the interfacial bond [108]. Therefore, the properties of the various materials, the content of the gradient components, and the processing parameters are essential elements in the design of multi-material functional gradient structures.

4. Challenges and Prospects for Hybrid Laser Additive Manufacturing

Hybrid laser additive manufacturing technology is a promising strategy for addressing laser additive manufacturing issues. Nonetheless, the technology faces a plethora of challenges that restrict its industrial application.

For starters, hybrid laser additive manufacturing technology places new requirements on materials. Because additive components can be employed for subtractive cutting using subtractive manufacturing technologies, the material's machinability must be considered. Incrementally manufactured parts, for example, are frequently harder than traditionally machined parts, putting a greater strain on the material reduction process. As a result, special materials for hybrid laser additive manufacturing should receive adequate attention.

Furthermore, multi-process, multi-energy field, and multi-material compounding are all part of hybrid laser additive manufacturing technology. The mechanism of interaction between various processes, energy fields, and materials is not yet apparent. The interface and synergistic control of multiple processes, the reasonable matching of process parameters, the application of energy fields, and the performance control of multi-material transition zones are all issues that must be addressed. It is emphasized that the online monitoring and numerical simulation of hybrid additive manufacturing processes will be valuable in resolving such incompatible issues.

Ultimately, residual stresses are unavoidable in the metal laser additive manufacturing process, which has received considerable attention from academics because it has a significant impact on manufacturing accuracy and performance. There has been little research on the mechanism and control of residual stresses in hybrid laser additive manufacturing, which will become a major research focus in the future.

5. Conclusions

Hybrid laser additive manufacturing (AM) represents a significant advancement in the field of advanced manufacturing technologies. Combining the additive manufacturing capabilities of laser-based processes with other traditional processing technologies, hybrid AM offers a compelling solution for the fabrication of complex and high-precision components.

In practice, hybrid additive manufacturing technology demonstrates several unique advantages. Firstly, it combines the strengths of various processing technologies in a single system, allowing for a more streamlined production workflow. This integration reduces the need for multiple setups and tool changes, leading to improved overall efficiency and reduced production time. Nevertheless, it can perform further precision machining on the part surface after laser additive manufacturing to enhance surface quality and dimensional accuracy. By processing the parts produced through laser additive manufacturing, their

mechanical properties, surface finish, and overall performance are improved. This capability can enhance the functionality and lifespan of the parts. Hybrid AM with higher design freedom can achieve complex geometries with intricate features and internal structures. It also enables the incorporation of multiple materials in a single build. This versatility opens new possibilities for creating customized components with enhanced mechanical, thermal, and chemical properties.

Despite the significant progress made in the field of hybrid AM, there are still challenges to overcome. The mechanism of interaction between various processes, energy fields, and materials is not yet apparent. The interface and synergistic control of multiple processes, the reasonable matching of process parameters, the application of energy fields, and the performance control of multi-material transition zones are all issues that must be addressed. It is emphasized that the online monitoring and numerical simulation of hybrid additive manufacturing processes will be valuable in resolving such incompatible issues.

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